

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of BSc (BT/MI/BC) Examination March 2018

Course code & Name BE 113: Scope of Microbiology

Date: **29/03/2018** Day: **Thursday** Time: **10:00 am To 10:30 pm** Maximum Marks: **20**

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

20

- Which is NOT true of the name *Bacillus subtilis*?
 - Bacillus* is the genus to which it belongs
 - subtilis* is the strain to which it belongs
 - It is a scientific name
 - The name is based on binomial nomenclature
- Which of the following microorganism is NOT involved nitrogen fixation?
 - Rhizobium*
 - E.coli*
 - Azotobactor*
 - Nostoc*
- Bacterium having a round shape is known as _____.
 - coccus
 - bacillus
 - spirillum
 - all of these
- Prokaryote consists of _____ ribosomal subunits.
 - 80s
 - 70s
 - 100s
 - 60s
- Teichoic acids are found on _____ bacteria.
 - Gram-positive
 - Gram- Negative
 - both
 - none of these

6. According to Whittaker's classification system, algae is belonging to kingdom _____.
(i) Myceteae (ii) Monera (iii) Protista (iv) Plantae
7. _____is incorporated in agitated vessels to prevent vortex.
(i) Baffles (ii) Sparger (iii) Impeller (iv) None of these
8. Which of the following technique is used for the sterilization of hormones?
(i) Autoclave (ii) Dry heat (iii) Filtration (iv) All of these
9. _____is the principal component of wood which is susceptible to biodeterioration.
(i) Cellulose (ii) Lignin (iii) (i) and (ii) both (iv) Casein
10. Study related to fungi is known as _____.
(i) Mycology (ii) Phycology
(iii) Immunology (iv) Virology
11. _____medium can prevent the certain bacterial growth while permitting others.
(i) Nutrient (ii) Differential (iii) Basal (iv) Selective
12. The potent antagonists *Trichoderma* is known to secrete cell wall lysing enzymes____.
(i) chitinase (ii) protease (iii) amylase (iv) all of these
13. Bacteria that utilise chemical compound as energy source is known as _____.
(i) autotrophs (ii) lithotrophs
(iii) Chemotrophs (iv) organotrophs
14. Which of following is important for the bacterial locomotion?
(i) Flagella (ii) Capsule (iii) Ribosome (iv) Inclusion granules
15. Which organism is used as bio-pesticides?
(i) *E. coli* (ii) *Bacillus subtilis*
(iii) *Aspergillus niger* (iv) *Bacillus thuringiensis*
16. Bacterial Genomic DNA is linear while eukaryotic Genomic DNA is circular.
☐True ☐False
17. The cell wall of gram-positive bacteria is thicker than gram-negative bacteria.
☐True ☐False
18. Organisms in the same order are more closely related than those in the same family.
☐True ☐False
19. Parasitism is a phenomenon where one organism consumes another organism.
☐True ☐False
20. Most vaccines stimulate the immune system by containing disease-causing germs (bacteria and viruses), or materials that germs produce.
☐True ☐False

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I Semester of BSc (BT/MI/BC) Examination March 2018

Course code & Name BE 113: Scope of Microbiology

Date: **29/03/2018** Day: **Thursday** Time: **10:30 am To 1:00 pm** Maximum Marks: **50**

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed (any Five) 20

1. Explain in brief about the four Koch postulates.
2. What is microbiology? Discuss various scope of microbiology.
3. Write a note on “binomial system of nomenclature”.
4. Write a note on “Bacterial reproduction”.
5. What is symbiosis? Explain it with at least two examples.
6. Write the advantages of bio-pesticides.
7. What is vaccine? How do vaccines work?

SECTION - II

Q-III Answer the following questions as directed (any Five) 20

1. Give the difference between gram-positive and gram-negative bacteria.
2. Define: antibiosis, antibiotic, saprophytism and autotrophs
3. What is culture media? Differentiate culture media based on the physical state.
4. What is fermentation? Explain the various application of fermentation process.
5. Explain in brief about Carl-Woese three kingdom classification system.
6. Write a note on “size and shape of bacteria”

Q-IV Answer the following questions as directed (any Two) 10

1. Discuss the various applications of microbiology in the field of Medical, Agricultural, Industrial and Food microbiology.
2. Draw the labeled diagram of bacterial cell and briefly describe the function of cytoplasm, and structures within it.
3. What is antagonism? Explain the different types of antagonistic interactions.